



01 · PROBLEM UNDERSTANDING

Real-Time Fraud Detection for Small Merchants

Problem Statement ID

Omni_FinTech_3

Track

FinTech

Small merchants using digital payments are increasingly targeted by fraud, often with little protection. FraudShield is a proposed real-time system to help detect and prevent payment fraud for small businesses.

Practical challenges merchants face today



Slow identification. Unusual transactions are difficult for a small team to spot quickly, especially in real time.



No dedicated fraud team. Most small merchants have nobody watching transactions full time.



Noisy rule-based alerts. Static rule engines tend to generate excessive false alerts, which merchants learn to ignore.



No "why." A binary fraud / not-fraud label gives no explanation the merchant can act on.



Financial impact. Every hour of delayed detection is a direct, avoidable loss for a small business.

Who has this problem



Small retail merchants



Local businesses



Small online sellers



Micro & small enterprises

Why existing tools fall short

- Expensive for a small merchant's budget
- Complex to configure without a technical team
- Limited explainability behind each alert
- Not built around the merchant's own behavior

Why this matters now



Digital payment adoption among small merchants keeps growing, and fraud exposure grows with it.



Fraud tactics evolve quickly — static, one-size-fits-all rules fall behind fast.



For a thin-margin small business, even one large fraudulent transaction can be a serious setback.



Small merchants need **affordable, real-time, explainable** fraud protection that helps them make the right decision before a suspicious transaction becomes a financial loss.

02 Proposed Solution

FraudShield

An explainable, real-time fraud intelligence platform built specifically for small merchants — it does not just flag a transaction, it tells the merchant why, and what to do next.



01 Detect

Analyze every transaction in real time using transaction-level and behavioral signals as it happens.

Velocity

Amount

Device / session

Timing



02 Explain

Surface the specific factors contributing to the risk score — not just a label.

91 / 100

HIGH RISK

SIMULATED EXAMPLE — NOT A MEASURED RESULT

- Amount well above merchant's normal range
- Unusual transaction time
- Multiple failed attempts beforehand
- New device / session



03 Act

Translate the score into one clear, recommended action for the merchant to take.

APPROVE

REVIEW / VERIFY

BLOCK



04 Learn

Merchant feedback continuously refines detection and each merchant's behavioral baseline.

"Fraud"

"Legitimate"

"Reviewed"



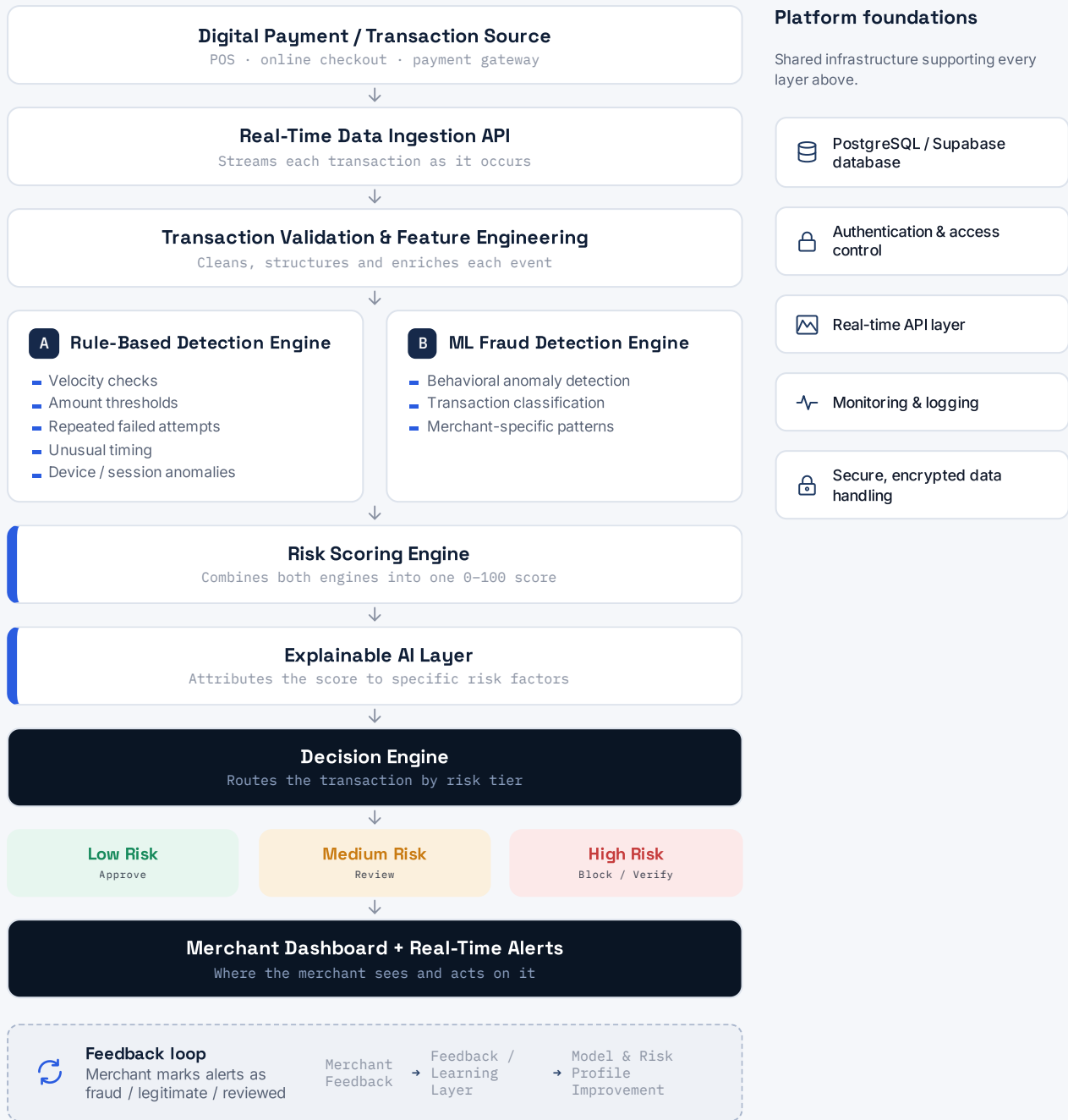
FraudShield combines behavioral anomaly detection, adaptive risk scoring, explainable alerts and merchant feedback into one lightweight fraud-prevention platform built for small businesses.

FraudShield surfaces risk and recommends action — it does not claim guaranteed fraud prevention or 100% detection accuracy.

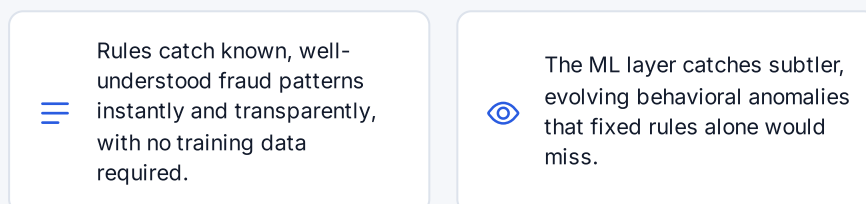


03 System Architecture

Every transaction moves through validation, a dual detection layer, adaptive scoring and an explainability pass before the merchant sees a recommendation — with feedback flowing back in to keep the model current.



Why a two-engine approach





04 Key Features

01 Real-Time Transaction Monitoring

Continuously watches transactions as they happen and flags suspicious activity with minimal delay.

02 Dynamic Fraud Risk Score

Produces one understandable 0–100 risk score per transaction, instead of a raw model output.

03 Explainable Fraud Alerts

Shows the contributing risk factors behind every alert, not just a "fraud detected" label.

04 Merchant Behavioral Baseline

Learns each merchant's normal transaction pattern and flags deviations from it specifically.

05 Smart Action Recommendations

Recommends Approve, Review, Verify or Block based on the transaction's current risk tier.

06 Feedback-Driven Improvement

Lets merchants mark alerts as legitimate or fraudulent so detection keeps improving over time.

07 Fraud Analytics Dashboard

One view of transaction volume, flagged transactions, fraud trends, risk distribution and daily / weekly patterns — with confirmed-fraud vs. false-positive feedback built in.

Transaction volume Flagged transactions Risk distribution Weekly patterns

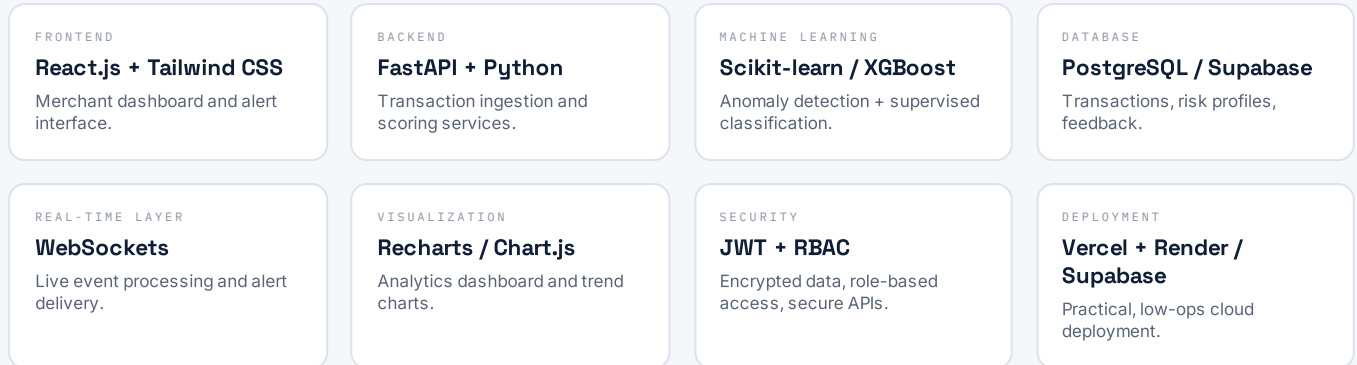
What the merchant sees



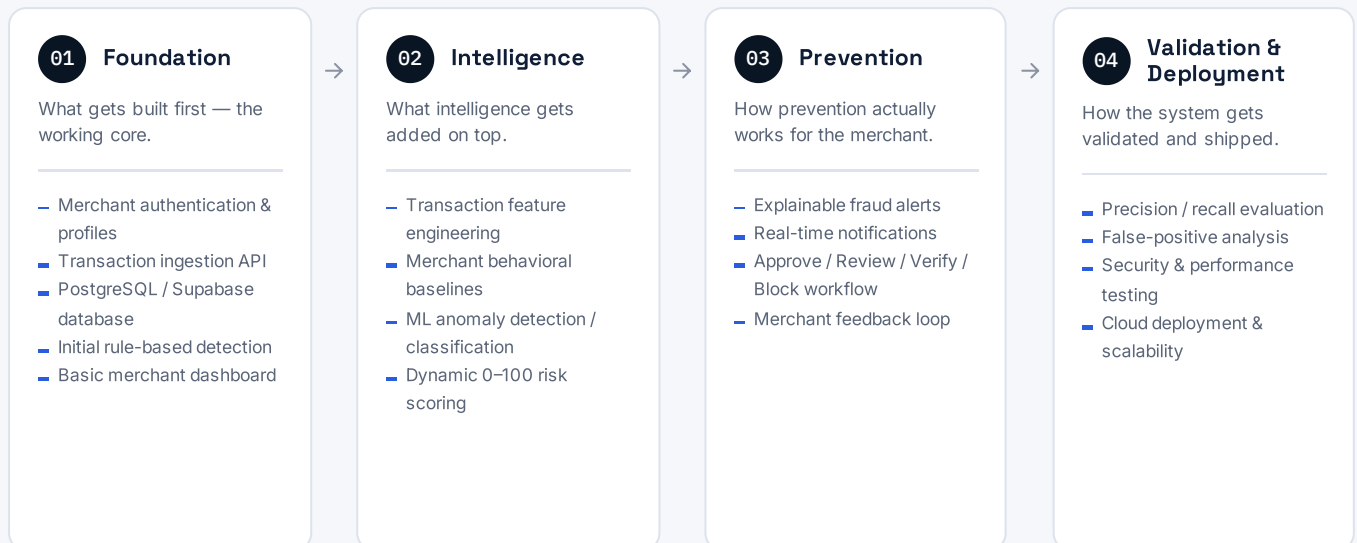


05 Technology Stack & Implementation Plan

Technology stack



Implementation roadmap



Expected Impact **Faster** detection **Fewer** losses **Explainable** decisions **Affordable** protection **Scalable** intelligence



Can this actually be built? Every layer of the stack — React, FastAPI, Scikit-learn, PostgreSQL — is widely adopted and well documented, so a small team can realistically ship the Phase 1 MVP within the hackathon build window.

FraudShield is a proposed hackathon solution and MVP implementation plan. No claims of a trained model, measured accuracy or live deployment are made at this stage.